

Programme : B. Sc. Information Technology (2026 - 27)		Semester V	
Course : Computational Indic Studies		Code : UICCIS301	
Teaching Scheme		Evaluation Scheme	
Lecture	Credits	Continuous Assessment (CA)	Semester End Examinations (SEE)
		(Marks - 20)	(Marks- 30)
02 Lab based	2	20	30
Internal Component (Practical Break up)			
Examination		Mini Project/Case study/ Field Visit (report to be submitted and certified prior to practical examination)	
30 Marks		20 Marks	
Learning Objectives			
CO1: To understand the evolution, structure, and linguistic features of major historical scripts used in Indian knowledge traditions.			
CO2: To analyze challenges in digitizing historical manuscripts and apply basic computational tools such as Unicode encoding and OCR for Indic scripts.			
CO3: To evaluate the role of Artificial Intelligence and Natural Language Processing in manuscript analysis, restoration, and script recognition.			
CO4: To explain the importance of digital technologies in preserving Indian knowledge systems and cultural heritage.			
CO5: To apply immersive technologies such as Augmented Reality (AR) and Virtual Reality (VR) to conceptualize digital heritage platforms.			
CO6: To design a prototype concept for an IKS Digital Museum integrating AI, AR, and VR for knowledge preservation and interactive learning.			
Learning Outcomes:			
After successful completion of the course, students will be able to:			
1. Describe the evolution and structural features of major scripts used in the Indian subcontinent.			
2. Explain key characteristics of Sanskrit, Pali, Arabic, and Gurmukhi manuscript traditions.			
3. Analyze challenges involved in digitizing historical manuscripts including script variation and document degradation.			
4. Apply basic digital tools such as Unicode encoding and OCR for processing Indic scripts.			
5. Explain the role of Artificial Intelligence, Augmented Reality, and Virtual Reality in preserving Indian knowledge systems.			
6. Design a conceptual prototype for an IKS Digital Museum using immersive technologies.			
Pedagogy :			
Lab based			

Module	Module Content	Duration of Module
I	Scriptology and Digital Manuscript Analysis (Lab-Oriented) Introduction to Scriptology and the evolution of writing systems in the Indian subcontinent. Study of major classical and medieval scripts including Sanskrit script traditions, Pali manuscript traditions, Arabic script in medieval scholarship, and Gurmukhi script. Structural characteristics of scripts including phonetic representation, writing patterns, and orthographic systems. Digitization challenges in historical manuscripts including script variations, document degradation, and multilingual complexity. Practical introduction to Unicode encoding, OCR technologies, and digital text processing for Indic scripts. Applications of Artificial Intelligence and Natural Language Processing for script recognition, manuscript restoration, and textual analysis.	15
II	Immersive Technologies for Preserving Indian Knowledge Digital heritage preservation using modern computing technologies. Introduction to Augmented Reality (AR) and Virtual Reality (VR) for cultural and knowledge preservation. Concept of IKS Virtual Museums: Digitally recreating ancient libraries, manuscripts, and knowledge systems using immersive technologies. Applications such as: • VR reconstruction of ancient universities and knowledge centers • AR-based interactive learning of scripts and manuscripts • 3D visualization of ancient instruments and knowledge artifacts Use of AI, AR, and VR to build interactive knowledge repositories. Case studies of digital heritage initiatives and computational platforms used to preserve cultural knowledge. Design concept: IKS Digital Museum Prototype where students conceptualize immersive exhibits of Indian knowledge traditions.	15

Books and References:

SrNo	Dataset/Books	Authors	Publisher	Date
1	Datasets Indian Scripts		Kaggle	
2	Word Level Script Identification Using Convolutional Neural Network Enhancement for Scenic Images ACM Transactions on Asian and Low-Resource Language Information Processing 21(4):1-29 DOI:10.1145/3506699			March 2022
3	Synthetically Created Dataset Historic Scripts to Modern Vision: A Novel Dataset and A VLM Framework for Transliteration of Modi Script to Devanagari		https://aikosh.indiaai.gov.in/home/datasets/details/modescript_synthetic_dataset.html	2025
4	Introduction to Indian Knowledge System: Concepts and Applications	B. Mahadevan, Vinayak Rajat Bhat & Nagendra Pavithran	PHI Learning Covers scientific traditions, mathematics, knowledge organization, and modern applications of IKS.	2022